Modular Framework for Harmful Meme Interpretation

Integrated Implementation Specification
Internal Evidence Extractor · External Knowledge Acquisition · Knowledge Relevance Filter/Verifier · Evidence Fusion & Reasoning · Structured Interpretation Head
Version 1.0

Document Purpose

This document serves as an implementation guideline that consolidates the five modules discussed earlier into a single integrated specification. It consistently organizes each module's role, inputs and outputs, recommended backbones, dimensional design, key equations, training losses, and inter-module connections.

1. Overall Framework Overview

Overall flow: Input meme (image + OCR text) -> Internal Evidence Bank -> Knowledge Candidates Bank -> Verified Knowledge Bank -> Shared/Task-aware Reasoning State -> Structured JSON Output

- Stage A structures evidence that can be directly observed inside the meme.
- Stage B retrieves external knowledge candidates and forms short hypotheses based on internal evidence.
- Stage C filters the retrieved results and retains only the knowledge that is actually valid for interpreting the current sample.
- Stage D performs sample-specific fusion by using internal evidence as the query and verified knowledge as the key/value.
- Stage E outputs harmfulness, target, intent, tactic, supporting evidence, and rationale in a structured format.

1.1 Common Design Principles

- Common hidden dim: All module output tokens are projected into a 256-dimensional space.
- Retrieval principle: Maintain the principle of retrieve first, generate later, verify always.
- Reasoning principle: Pass only verified knowledge, not raw retrieval results, to Stage D.
- Output principle: Stabilize structured labels and evidence attribution before free-form generation.
- Training principle: Use staged training by first stabilizing A -> B -> C individually, then connecting D/E.

1.2 Common Dimensions and Basic Symbols

- Input image: $I \in R^{3 \times 224 \times 224}$
- Number of internal evidence tokens: $N_e \approx 12$
- Number of verified knowledge tokens: $N_k \approx 8$
- Common hidden dimension: $d = 256$
- Stage D output latent vectors: r_{shared}, h_{target}, h_{intent}, $h_{tactic} \in R^{256}$

2. Stage A - Internal Evidence Extractor

Role: This stage structures directly observable global visual cues, OCR semantic cues, local symbols/ROIs, and image-text incongruity within the meme to create the Internal Evidence Bank.

2.1 Inputs / Outputs

- Inputs: sample_id, dataset_name, image, OCR text, raw label, optional metadata
- Outputs: evidence_tokens [N_e, 256], ROI metadata, text spans, multimodal relation / rhetorical cue / knowledge_need auxiliary scores

2.2 Submodule Composition

- Global Visual Encoder: OpenCLIP ViT-L/14 recommended. Extracts a global image embedding and patch tokens.
- Text Semantic Encoder: DeBERTa-v3-base or XLM-R-base recommended. Encodes the full OCR text into contextual tokens and a global text embedding.
- Local Object / Symbol Extractor: YOLO-World-L or GroundingDINO recommended. Uses generic / OCR-guided / metadata-guided prompts.
- Cross-modal Incongruity Analyzer: Extracts irony / incongruity cues through bidirectional cross-attention between image patch queries and text token key/values.

2.3 Key Equations

- Visual projector: V_hat = V W_v + b_v, g_v_hat = g_v W_g + b_g
- Text projector: H_hat = H W_t + b_t, g_t_hat = g_t W_gt + b_tg
- Cross-modal incongruity: Z_v->t = MHA(Q = V_hat, K = H_hat, V = H_hat), Z_t->v = MHA(Q = H_hat, K = V_hat, V = V_hat)
- Incongruity vector: h_inc = MLP([pool(Z_v->t); pool(Z_t->v); g_v_hat; g_t_hat; g_v_hat ⊙ g_t_hat; |g_v_hat - g_t_hat|])

2.4 Representative Output Format

- global_visual_token [256]
- global_text_token [256]
- roi_tokens [K, 256]
- incongruity_token [256]
- evidence_tokens [N_e, 256]
- scores: target_presence, multimodal_relation, rhetorical_cues, knowledge_need

2.5 Main Losses

- L_global_align: image-text alignment loss
- L_roi_det: open-vocabulary region proposal / selection loss
- L_incongruity: binary or ranking loss
- L_tp / L_mr / L_rh / L_need: auxiliary losses for target presence, multimodal relation, rhetorical cues, and knowledge need

3. Stage B - External Knowledge Acquisition

Role: Based on the Internal Evidence Bank, this stage constructs retrieval-oriented queries, links entities/concepts, and builds the Knowledge Candidates Bank through multi-source retrieval and limited context generation.

3.1 Inputs / Outputs

- Inputs: OCR text, target/evidence spans, ROI metadata, knowledge_need score, metadata
- Outputs: query_bundle, linked_nodes, retrieved_docs, knowledge_candidate_tokens [M, 256], hypothesis_tokens [H, 256]

3.2 Query Constructor
- Entity query: centered on people, organizations, groups, and proper nouns
- Event query: centered on current affairs / politics / social events
- Meme-template query: for searching characters, panel structure, and template meaning
- Social-context query: for interpreting prejudice, political criticism, and cultural codes
- Target-hypothesis query: auxiliary recall when an implicit target is suspected

3.3 Entity / Concept Linking
- Target sources: Wikipedia, Wikidata, ConceptNet, meme template lexicon
- Procedure: alias expansion -> bi-encoder retrieval -> cross-encoder rerank
- Recommended models: bge-base/e5-base + DeBERTa-v3-base cross-encoder

3.4 Hybrid Retrieval
- Index design: general knowledge / meme culture / event source
- Sparse retrieval: Pyserini BM25
- Dense retrieval: dual encoder + FAISS
- Fusion: RRF or normalized weighted sum

3.5 Context Augmentation Generator
- Uses only top-k retrieved evidence, not the raw meme, as input.
- Generates short contextual summary / target hypothesis / intent hypothesis / tactic hypothesis, not long explanations.
- Recommended model: a constrained generator at the level of Flan-T5-base.

3.6 Main Losses
- L_link: entity linking loss
- L_ret: dense retrieval contrastive loss
- L_ctx: constrained seq2seq hypothesis generation loss
- L_qroute: query type routing supervision

4. Stage C - Knowledge Relevance Filter / Verifier
Role: Among the retrieved knowledge candidates, this stage keeps only the documents that are truly relevant, supportable, temporally/culturally/source-valid, and not overly redundant for the current meme interpretation, thereby creating the Verified Knowledge Bank.

4.1 Inputs / Outputs
- Inputs: internal evidence summary q_int, retrieved docs, knowledge_candidate_tokens [M, 256], generated hypotheses, metadata
- Outputs: verified_tokens [K_ver, 256], verified_docs, support_matrix [K_ver, 3], final verification scores

4.2 Evidence-aware Relevance Scorer
- q_int = MLP([g_v_hat; g_t_hat; h_inc; Pool(ROI)]) in R^256
- Candidate relevance feature: [q_int; k_i; q_int ⊙ k_i; |q_int - k_i|]
- Output: s_rel(i) in [0, 1]

4.3 Consistency / Support Verifier
- Constructs atomic claims for target / intent / tactic.

- Performs 3-way NLI classification - support / contradict / insufficient - for each claim and candidate document pair.

- Also checks candidate-to-candidate consistency to reduce conflicts among top-ranked documents.

4.4 Credibility / Temporal / Cultural Validator

- Credibility prior: based on source domain / archive quality
- Temporal compatibility: based on the time difference between the sample and the document
- Cultural compatibility: based on language / country / region / platform alignment

4.5 Redundancy Reduction

- Computes cosine similarity among verified candidate embeddings.
- Selects cluster-aware top-k after semantic clustering.
- Final score is a weighted sum of relevance + support + validity.

4.6 Main Losses

- L_rel: relevance BCE/CE
- L_sup: support/contradict/insufficient CE
- L_rank: positive > negative ranking loss
- L_cal: confidence calibration loss (optional)

5. Stage D - Evidence Fusion & Reasoning

Role: This stage performs sample-specific cross-attention fusion using the Internal Evidence Bank as query and the Verified Knowledge Bank as key/value, then controls internal/external dependency through gating to generate reasoning latents for target / intent / tactic.

5.1 Inputs / Outputs

- Inputs: internal evidence tokens [N_e, 256], verified knowledge tokens [N_k, 256], support matrix, auxiliary priors

- Outputs: shared_reasoning_state [256], target_latent [256], intent_latent [256], tactic_latent [256], fused_internal_tokens [N_e, 256]

5.2 Internal Evidence Aggregator

- Adds source-type embedding and dataset embedding to tokens.
- Uses a 2-layer Transformer encoder to organize interactions among internal tokens.
- Computes q_int or a shared internal summary through attention pooling.

5.3 Knowledge-conditioned Cross-attention

- Q = internal memory, K/V = verified knowledge memory
- Applies standard multi-head cross-attention followed by residual connection + FFN
- A support-aware bias can be added to attention logits

5.4 Gating

- Token-level gate: determines how much each internal token accepts external fusion
- Sample-level gate: adjusts the overall external injection ratio based on knowledge_need
- Head-level gate: reflects different degrees of external dependency for target / intent / tactic

5.5 Task-aware Reasoning Heads

- Branches target / intent / tactic-specific latents from the shared reasoning state.

- Uses a shared + private MLP structure to reduce multi-task interference.
- Tactic depends more on internal incongruity, while target depends relatively more on external knowledge.

5.6 Main Losses

- L_task: downstream prediction loss for target / intent / tactic
- L_suff: evidence sufficiency regularizer
- L_cons: inter-task consistency regularizer
- L_distill: optional teacher reasoning distillation

6. Stage E - Structured Interpretation Head

Role: This stage converts fused reasoning latents into structured outputs for harmfulness, target, intent, and tactic, selects supporting evidence, and generates evidence-grounded rationale.

6.1 Inputs / Outputs

- Inputs: shared_reasoning_state, target / intent / tactic latent, fused internal tokens, verified tokens, attribution metadata
- Outputs: JSON-like structured output including harmfulness, target, intent, tactic, supporting_evidence, and rationale

6.2 Harmfulness Head

- Input: [r_shared; h_target; h_intent; h_tactic] in R^1024
- Output: harmful / non-harmful score
- Uses binary CE or sigmoid BCE

6.3 Target / Intent / Tactic Heads

- Target: presence (explicit / implicit / absent), granularity, protected attribute multi-label
- Intent: primary intent, stance, secondary intent, background knowledge dependency
- Tactic: rhetorical tactic multi-label, multimodal relation, template/structure tactic

6.4 Evidence Attribution Layer

- Internal evidence selector: selects evidence tokens from OCR spans, ROI, and cross-modal cues
- External evidence selector: selects top-k supporting documents from verified knowledge docs
- Does not use attention weights directly as explanations; instead, it uses a separate selection head.

6.5 Rationale Generator

- Recommends a template-based rationale for the MVP
- Input slots: text evidence, visual evidence, external evidence, predicted target / intent / tactic
- The extended version can evolve into constrained seq2seq rationale generation.

6.6 Main Losses

- L_harm, L_target, L_intent, L_tactic
- L_evid: internal / external evidence selection loss
- L_comp: label compatibility constraint
- L_rat: optional rationale generation loss

7. Integrated Input/Output Summary by Module

- Stage A: image, OCR text -> Internal Evidence Bank -> N_e x 256
- Stage B: Internal Evidence Bank -> Knowledge Candidates Bank -> M x 256
- Stage C: Internal Bank + Candidates -> Verified Knowledge Bank -> K_ver x 256
- Stage D: Internal Bank + Verified Bank -> shared/task-aware reasoning state -> 256 / token memory
- Stage E: reasoning latent + evidence banks -> Structured JSON output -> label logits + evidence ids

8. Recommended Class Structure

- Stage A: GlobalVisualEncoder, TextSemanticEncoder, LocalSymbolExtractor, CrossModalIncongruityAnalyzer, InternalEvidenceExtractor
- Stage B: QueryConstructor, EntityConceptLinker, HybridRetriever, ContextAugmentationGenerator, ExternalKnowledgeAcquisition
- Stage C: EvidenceAwareRelevanceScorer, SupportContradictionVerifier, CredibilityTemporalValidator, RedundancyReducer, KnowledgeRelevanceFilterVerifier
- Stage D: InternalEvidenceAggregator, KnowledgeConditionedCrossAttention, EvidenceGate, TaskAwareReasoningHeads, EvidenceFusionReasoning
- Stage E: HarmfulnessHead, TargetHead, IntentHead, TacticHead, EvidenceAttributionLayer, TemplateRationaleGenerator, StructuredInterpretationHead

9. Integrated Training Order

- Step 1: Freeze Stage A and first stabilize OCR / ROI / incongruity quality.
- Step 2: Train Stage B's query / linking / retrieval separately and inspect index quality.
- Step 3: Train the Stage C verifier sufficiently to secure strong wrong-knowledge suppression capability.
- Step 4: Connect Stage D and Stage E for end-task fine-tuning.
- Step 5: Add rationale generation at the end; prioritize structured output + evidence attribution in the early stage.

10. Final Recommended MVP Configuration

- Extractor: OpenCLIP ViT-L/14 + DeBERTa-v3-base + YOLO-World-L + bidirectional incongruity attention
- Retrieval: Pyserini BM25 + dense dual encoder + FAISS + lightweight context generator
- Verifier: relevance MLP + DeBERTa-v3-base support verifier + temporal/source prior + semantic clustering
- Fusion: 256d token memory + 1-hop cross-attention + token/sample/head-level gating
- Head: harmfulness + target + intent + tactic structured heads + evidence attribution + template rationale

11. Conclusion

Key takeaway: The success of this framework does not lie in using a larger monolithic model. Rather, the core of stable harmful meme interpretation lies in the structuring of OCR and symbolic cues, generation that does not precede retrieval, strong verification after retrieval, sample-specific fusion, and faithful structured output.